BLUESTOCK MUTUAL FUND ANALYTICS

A Unified Quantitative Evaluation, Data Cleaning Pipeline & Portfolio Optimisation Framework

Deliverable Reference: D7 - Final Capstone Technical Report

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Platform Config: Python, SQLite, ReportLab, Streamlit, Git

GitHub Target: MUTUALFUNDSDATAANALYTICS

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1. Executive Summary

This capstone technical report details the build, calculations, and analytical insights generated for the Bluestock Mutual Fund Capstone Project. The system coordinates a fully automated weekday ETL data pipeline pulling live mutual fund Net Asset Values (NAV) from the open-source AMFI API and staging it inside an optimized SQLite relational schema. Trailing returns, risk metrics, and investor transaction statistics are compiled programmatically. Finally, a responsive web dashboard deployed in Streamlit provides interactive data slicing, automated recommendations, and advanced portfolio optimization scripts for retail investors.

2. Database Schema & Data Models

The project database is stored in **data/db/bluestock_mf.db**. It utilizes five main relational tables designed to ensure 3NF integrity:

• **dim_fund**: Master dimension table containing AMFI scheme codes, fund houses, categories, benchmarks, and plan structures.

• **fact_nav**: Historical daily Net Asset Value database, containing scheme code mapping and reindexed dates.

• **fact_transactions**: Investor transaction ledger cataloging transactional sizes, state, KYC, age, and payment types.

• **fact_performance**: Compiled analytical metrics tracking trailing return percentages, Alpha, Beta, Sharpe, and Drawdowns.

• **portfolio_holdings**: Fund holdings weights and sectors utilized to check portfolio diversification levels.

3. ETL Pipeline & Reindexing Methodology

The ETL pipeline script, located in **scripts/etl_pipeline.py**, runs without manual intervention. It fetches the latest daily NAV updates from *api.mfapi.in* for direct and regular plans. A critical phase in mutual fund analytics is the handling of dates where NAV is not published (weekends and public holidays). Common analytics mistakes include skipping holiday dates, which biases return time-series, or reindexing to calendar days without forward-filling.

Reindexing & ffill() Rules Applied:

1. For each fund, the script identifies the minimum and maximum NAV date.

2. It reindexes the time series to a continuous daily date index (calendar range including Saturday and Sunday).

3. It applies a forward-fill (ffill()) to carry Friday NAVs over the weekend.

4. Returns and CAGRs are annualized based on standard trading days: **252 business days** per calendar year, preventing return bias.

4. Performance Rankings Scorecard

The table below represents the top 8 mutual fund schemes sorted by 3-Year CAGR returns. Data is derived programmatically from the SQLite database:

Scheme Name	Category	3Yr CAGR	Sharpe	Risk Grade
SBI Bluechip Fund - Regular Plan - Growth	Large Cap	28.56%	0.96	Moderate
Mirae Asset Large Cap Fund - Regular - Growth	Large Cap	25.13%	1.20	Moderate
DSP Midcap Fund - Regular - Growth	Mid Cap	24.69%	0.90	High
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	24.41%	0.87	High
ABSL Frontline Equity Fund - Regular - Growth	Large Cap	22.40%	0.79	Moderate
Kotak Flexicap Fund - Regular - Growth	Flexi Cap	21.67%	1.07	Moderately High
Axis Midcap Fund - Regular - Growth	Mid Cap	21.29%	0.77	High
Mirae Asset Tax Saver Fund - Regular - Growth	ELSS	20.89%	1.01	High

5. Advanced Financial Engineering & Risk Metrics

For deep risk analysis, we calculate **95% Historical Value at Risk (VaR)** and **Conditional Value at Risk (CVaR)**. VaR represents the boundary limit below which daily returns fall with a 5% probability. CVaR computes the expected average loss given that the return exceeds the VaR threshold (the mean of the 5% tail). These metrics help investors evaluate downside tail risks during market stress events.

Sector Concentration Analysis (Herfindahl-Hirschman Index - HHI):

HHI values are computed by squaring sector weights: HHI = sum(Sector_Weight_pct ^ 2). This metric is calculated for all equity funds. A low HHI (below 1,200) denotes well-diversified portfolios (e.g. diversified large-cap funds). A high HHI (above 2,000) shows thematic or sector-focused funds, representing concentration risks.

Top Volatile Funds & Risk Scores:

Scheme Name	Category	Annualised Volatility	Morningstar Rating	Risk Grade

6. Investor Cohort & Retention Analysis

We perform demographic cohort splits and retention flags using the transaction ledger. **Cohort analysis** segments investors based on their first transaction year (2024 vs 2025). **SIP Continuity analysis** tracks accounts with 6 or more monthly transactions. If the gap between transactions exceeds 35 days, the account is flagged as 'At Risk' of churning. These flagged customer records are stored inside data/processed/sip_continuity.csv for targeting customer retention email campaigns.